In [1]:
```
!pip install gensim
```

```
Requirement already satisfied: gensim in C:\Users\manis\AppData\Local\Python\pyth
oncore-3.12-64\Lib\site-packages (4.4.0)
Requirement already satisfied: numpy>=1.18.5 in C:\Users\manis\AppData\Local\Pyth
on\pythoncore-3.12-64\Lib\site-packages (from gensim) (2.5.4)
Requirement already satisfied: scipy>=1.7.0 in C:\Users\manis\AppData\Local\Pytho
n\pythoncore-3.12-64\Lib\site-packages (from gensim) (1.18.1)
Requirement already satisfied: smart_open>=1.8.1 in C:\Users\manis\AppData\Local
\Python\pythoncore-3.12-64\Lib\site-packages (from gensim) (9.0.1)
Requirement already satisfied: wrapt in C:\Users\manis\AppData\Local\Python\pytho
ncore-3.12-64\Lib\site-packages (from smart_open>=1.8.1->gensim) (2.5.2)
```

In [4]:
```
# a. Import necessary packages
import pandas as pd
import numpy as np
import re
from gensim.models import Word2Vec

# Load Twitter dataset
data = pd.read_csv('twitter_dataset.csv')

# Display first five rows
print(data.head())

# Display column names
print(data.columns)

# Select tweet column
tweets = data['Text'].dropna().astype(str)

# Clean and tokenize tweets
def clean_text(text):
    text = text.lower()
    text = re.sub(r'http\S+|www\S+', '', text)
    text = re.sub(r'[^a-z\s]', '', text)
    words = text.split()
    return words

sentences = [clean_text(tweet) for tweet in tweets]

# Remove empty sentences
sentences = [s for s in sentences if len(s) > 0]

print("Total tweets:", len(sentences))
print("Sample tokens:", sentences[0])

# b. Train Word2Vec CBOW model
model = Word2Vec(
    sentences=sentences,
    vector_size=100,
    window=5,
    min_count=2,
    sg=0,
    workers=4,
    epochs=10
)

# Save the trained model
```

```
model.save("twitter_cbow.model")

print("CBOW model training completed!")

# c. Evaluate the results

# Display vocabulary
print("Vocabulary size:", len(model.wv))

# Display word embedding
word = list(model.wv.index_to_key)[0]

print("Selected word:", word)
print("Word embedding:", model.wv[word])

# Find similar words
print("Similar words:")
print(model.wv.most_similar(word, topn=5))
```

```
   Tweet_ID        Username  \
0         1         julie81
1         2   richardhester
2         3  williamsjoseph
3         4     danielsmary
4         5      carlwarren

                                                Text  Retweets  Likes  \
0  Party least receive say or single. Prevent pre...         2     25
1  Hotel still Congress may member staff. Media d...        35     29
2  Nice be her debate industry that year. Film wh...        51     25
3  Laugh explain situation career occur serious. ...        37     18
4  Involve sense former often approach government...        27     80

             Timestamp
0  2023-01-30 11:00:51
1  2023-01-02 22:45:58
2  2023-01-18 11:25:19
3  2023-04-10 22:06:29
4  2023-01-24 07:12:21
Index(['Tweet_ID', 'Username', 'Text', 'Retweets', 'Likes', 'Timestamp'],
      dtype='str')
Total tweets: 10000
Sample tokens: ['party', 'least', 'receive', 'say', 'or', 'single', 'prevent', 'p
revent', 'husband', 'affect', 'may', 'himself', 'cup', 'style', 'evening', 'prote
ct', 'effect', 'another', 'themselves', 'stage', 'perform', 'possible', 'try', 't
ax', 'share', 'style', 'television', 'with', 'successful', 'much', 'sell', 'devel
opment', 'economy', 'effect']
CBOW model training completed!
Vocabulary size: 971
Selected word: hard
Word embedding: [-1.50166854e-01  1.67672083e-01  1.00677371e-01  5.23522049e-02
 -1.73302589e-03 -2.86288649e-01 -3.68465036e-02  3.02786142e-01
 -3.80873773e-03  6.83507696e-02 -2.81090289e-01 -2.02634454e-01
 -4.96684350e-02  2.62074508e-02  1.78417824e-02 -2.25759984e-04
  5.56101352e-02 -4.20505814e-02  9.12992656e-02 -2.09978223e-01
  1.23192295e-01  8.58467370e-02  4.82841805e-02 -3.29516409e-03
 -6.29321113e-02  8.80872458e-02 -1.36895999e-01 -9.11460519e-02
  6.62374943e-02 -4.19884101e-02  1.34421498e-01  4.89645964e-03
 -5.39440326e-02  1.66116402e-01 -9.17382985e-02 -1.05995247e-02
 -8.94518718e-02 -2.34786019e-01 -1.44848883e-01 -2.54892379e-01
 -3.21175344e-02  5.03910482e-02  3.13184038e-02 -6.15554042e-02
 -8.88732448e-02 -2.26977337e-02 -1.24310181e-01  1.31564528e-01
  1.01446815e-01  1.38358012e-01 -3.33377868e-02 -1.43326893e-01
 -1.08754724e-01  7.02113360e-02 -2.20240489e-01  1.24496818e-01
  1.39819816e-01 -9.18248203e-03 -1.85363263e-01 -1.08171487e-02
  1.30016580e-01  1.90796375e-01 -2.46079952e-01 -6.38091937e-02
 -1.74463317e-01  3.73291709e-02 -4.75665629e-02 -1.30531769e-02
  3.64715373e-03  4.42540273e-02 -8.73037875e-02 -9.74260792e-02
  1.98914304e-01 -1.50363564e-01  1.14369236e-01  4.56441268e-02
 -5.63476272e-02 -9.01099592e-02 -1.61641374e-01  6.98653907e-02
 -6.07893057e-02 -1.00672640e-01  1.81118771e-02  2.62992948e-01
  1.36072278e-01  5.26164919e-02 -2.54047930e-01  2.15970963e-01
  1.32793829e-01  1.59384623e-01  4.71523032e-02 -3.32598388e-02
 -3.94212082e-02 -1.68876536e-02  1.77520245e-01  1.13603555e-01
  1.00726545e-01 -3.15840513e-01  1.18118025e-01 -1.67913847e-02]
Similar words:
[('result', 0.9705682396888733), ('perhaps', 0.9700422286987305), ('late', 0.9691
518545150757), ('describe', 0.9682878851890564), ('skin', 0.9682774543762207)]
```

In [ ]: